

AI-Powered Collections Strategy

Leveraging Agentic AI for Scalable, Fair, and
Effective Debt Management at Geldium

*Prepared by Aisha O. Inuwa | Task 4 Executive
Briefing*



Inspiring and
empowering
future professionals

Aisha Inuwa

GenAI Powered Data Analytics Job Simulation

Certificate of Completion
August 21st, 2026

Over the period of August 2026, Aisha Inuwa has completed practical tasks in:

Exploratory data analysis and risk profiling
Predicting delinquency with AI
Business report and data storytelling for collections strategy
Implementing an AI-driven collections strategy

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How the System Works: End-to-End Flow

01

DATA INPUTS

- Payment history (Month_1–Month_6)
- Credit utilization
- Credit score
- Loan balance & income

02

RISK SCORING & DECISION LOGIC

- Logistic regression risk probability
- + business rules (e.g. 2+ missed pmts)
- → Low / Medium / High risk tier

03

ACTIONS & OUTREACH

- Low/Med: automated reminders (SMS/email)
- High: flagged to Collections watch-list
- Human review for hardship offers

04

LEARNING LOOP

- Track cure & delinquency outcomes
- Monitor fairness metrics monthly
- Retrain & re-validate quarterly

How the System Works: Inputs, Logic & the Learning Loop

INPUTS & DECISION LOGIC

- Top predictive signals: repeated missed/late payments (Month_1–Month_6), credit utilization, credit score, loan balance, income
- A logistic regression model converts these into a delinquency probability (0–1) — chosen for transparency over more complex, harder-to-explain models
- Business rule layer: 2+ missed/late payments in 6 months → added to weekly watch-list
- Probability + rules together set a Low / Medium / High risk tier per customer

ACTIONS & THE LEARNING LOOP

- Low risk: no action, standard statement only
- Medium risk: automated reminders and payment nudges (SMS/email)
- High risk (watch-list): routed to Collections staff for outreach and hardship/repayment options
- Learning loop: cure & delinquency rates tracked through a 90-day pilot; fairness metrics reviewed monthly; model retrained and re-validated quarterly

Role of Agentic AI

Routine, low-risk actions run autonomously; anything that changes a customer's financial standing or account status requires human sign-off.

Autonomous AI Actions	Human Oversight Required
Score & tier every account weekly (risk probability + business rules)	Approve or adjust risk-tier thresholds each quarter
Build the flagged watch-list (2+ missed/late payments in 6 months)	Review and prioritize which watch-list accounts get contacted first
Send standard reminders & nudges (SMS/email) to low/medium-risk accounts	Approve hardship plans, payment-plan restructuring, or settlements
Log the reason codes and feature drivers behind every decision	Decide on account escalation (agency referral, credit bureau reporting, legal action)
Monitor fairness metrics (precision/recall by group) and flag anomalies	Investigate flagged fairness anomalies and approve any model or threshold changes
Track outcomes and prepare monthly performance reports	Sign off on quarterly model retraining and redeployment

Responsible AI Guardrails

01 Fairness Monitoring & Bias Testing

Standardize inconsistent data categories (e.g., employment labels) and compare precision, recall, and false-positive/negative rates across customer groups before and after deployment; correct material gaps.

02 Explainability by Design

Logistic regression was chosen specifically for its transparency. Every flagged account carries a plain-language reason (e.g., “2 missed payments + high utilization”), not just a score.

03 Regulatory & Data Compliance

Actions align with fair-lending and debt-collection rules (e.g., ECOA, FDCPA) and data-privacy requirements. Model limitations and data lineage are documented and version-controlled.

04 Human-in-the-Loop & Reversibility

The model supports prioritization and outreach only. No irreversible action (credit bureau reporting, legal referral) is automated without human review, and performance is re-validated on a regular cadence.

Expected Business Impact

BUSINESS KPIs (Quantitative)

- Reduced delinquency rate among flagged watch-list accounts, measured against baseline during the 90-day pilot
- Lower cost-to-collect: routine reminders run automatically, freeing staff time for complex, high-risk cases
- Faster time-to-contact: weekly automated scoring replaces slower manual review cycles
- Higher cure rate: more accounts return to current status before reaching serious delinquency

CUSTOMER OUTCOMES (Qualitative)

- Earlier, more supportive contact — instead of contact only after an account is already seriously delinquent
- Personalized hardship and repayment options matched to a customer's actual situation
- More consistent, explainable treatment across customer groups, reducing the risk of unfair outcomes
- Greater transparency: customers understand why they were contacted and what their options are

Implementation Roadmap: 90-Day Pilot

PHASE 1

Weeks 1–2

Data Readiness

- Clean inconsistent employment categories
- Validate credit-utilization outliers (values above 100%)
- Finalize watch-list business rules and risk-tier thresholds

PHASE 2

Weeks 3–12

Pilot Launch

- Weekly automated scoring goes live
- Automated reminders + Collections outreach on watch-list accounts
- Fairness metrics monitored monthly during the pilot

PHASE 3

Week 13

Review & Decide

- Compare delinquency & cure rates vs. baseline
- Run a fairness and explainability audit
- Decide whether to scale, adjust, or stop the workflow